

FIT2081 - Assignment 3 HD Feature Description

IMPORTANT

- **ONLY** to be used by students attempting HD > HD++ Marks. By nature of these being “above and beyond” marks, assignments missing basic features already mentioned in the A3 Spec will not be eligible for these marks
 - No code snippets are required
 - Duplicate the second page if you have more than two features
 - Make sure to edit the File name (add your student id) and Page Header (with your details)
-

1. Feature Name

ResetDetailsScreen

2. Path to the Feature Class (if applicable)

(e.g. `com/fit2081/a3_app/features/PatientMonitor.kt`)

Leave blank if not related to a specific class/.kt file

`com.fit2081.ian_34423680.nutritrackpro_app.viewmodel.ResetDetailsViewModel.kt`

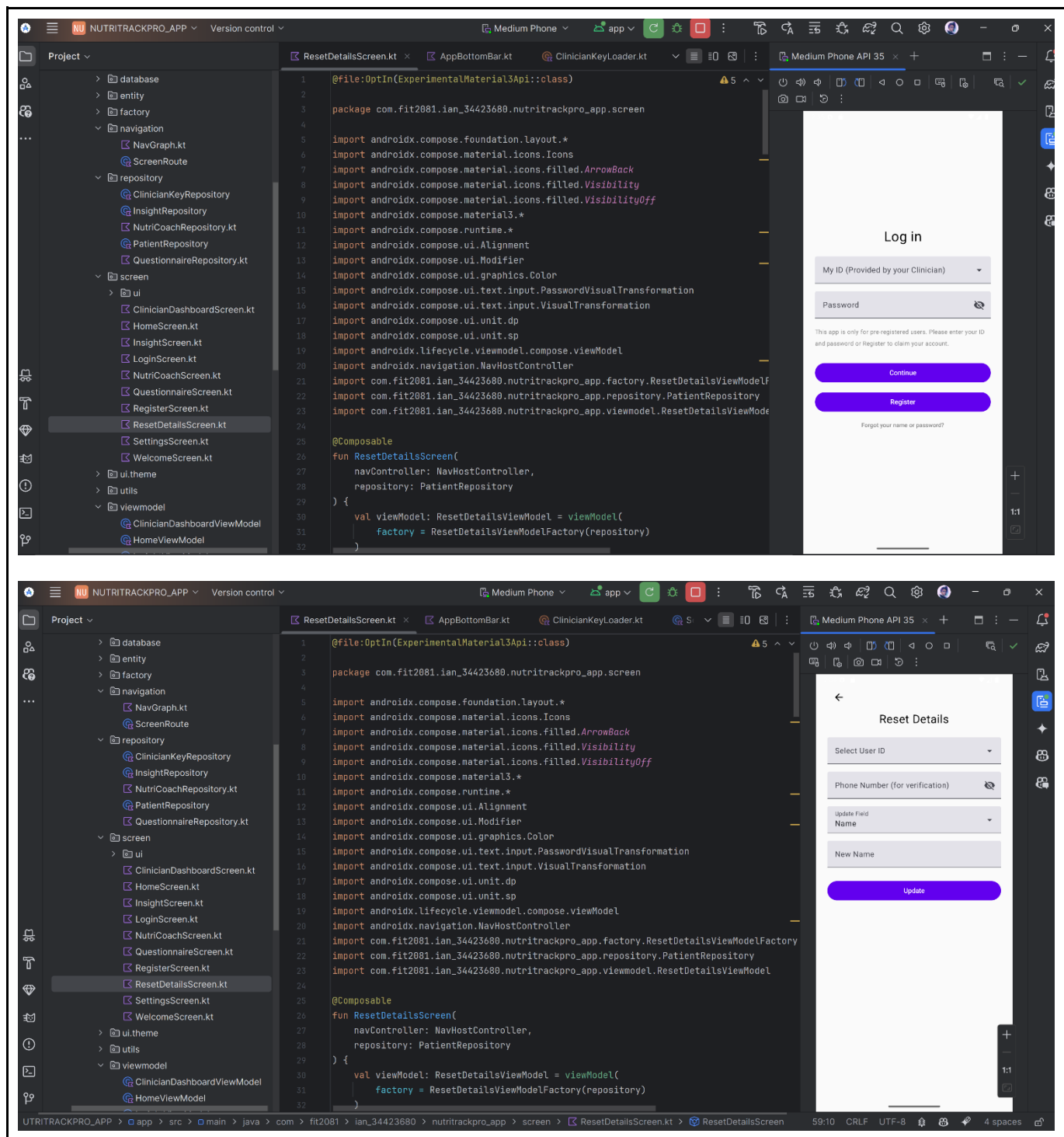
3. Feature Description (50-75 words)

- *The role of the feature and how it works*
- *Why/How this is going above and beyond the spec (i.e. how is this an “original” feature)*

This screen lets users reset either their name or password by first confirming their phone number for security. It includes live feedback for password strength and only allows updates if all conditions are met. It combines both reset options into one smooth flow, making it easier for users to recover their account details without needing admin help.

3. Feature Screenshots (max 2)

Name: [IAN LEONG ZHENG YAN] **Student ID:** [34423680] **Lab:** [TUTORIAL 06]



(If you have only one HD feature, leave this page blank.)

1. Feature Name

Change Name and Password in Settings Screen

2. Path to the Feature Class

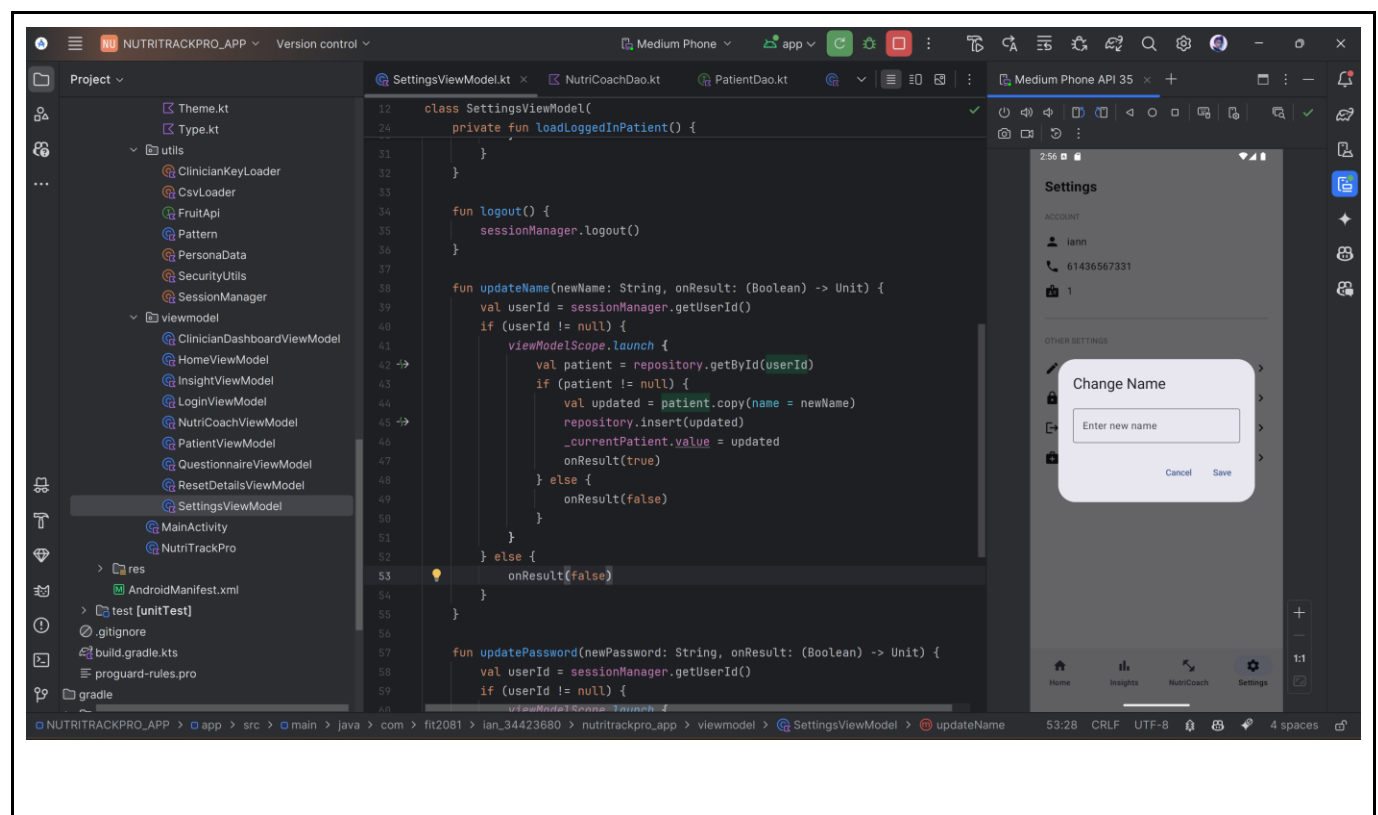
(e.g. `com.fit2081/a3_app/features/PatientMonitor.kt`)

`com.fit2081.ian_34423680.nutritrackpro_app.viewmodel.SettingsViewModel.kt`

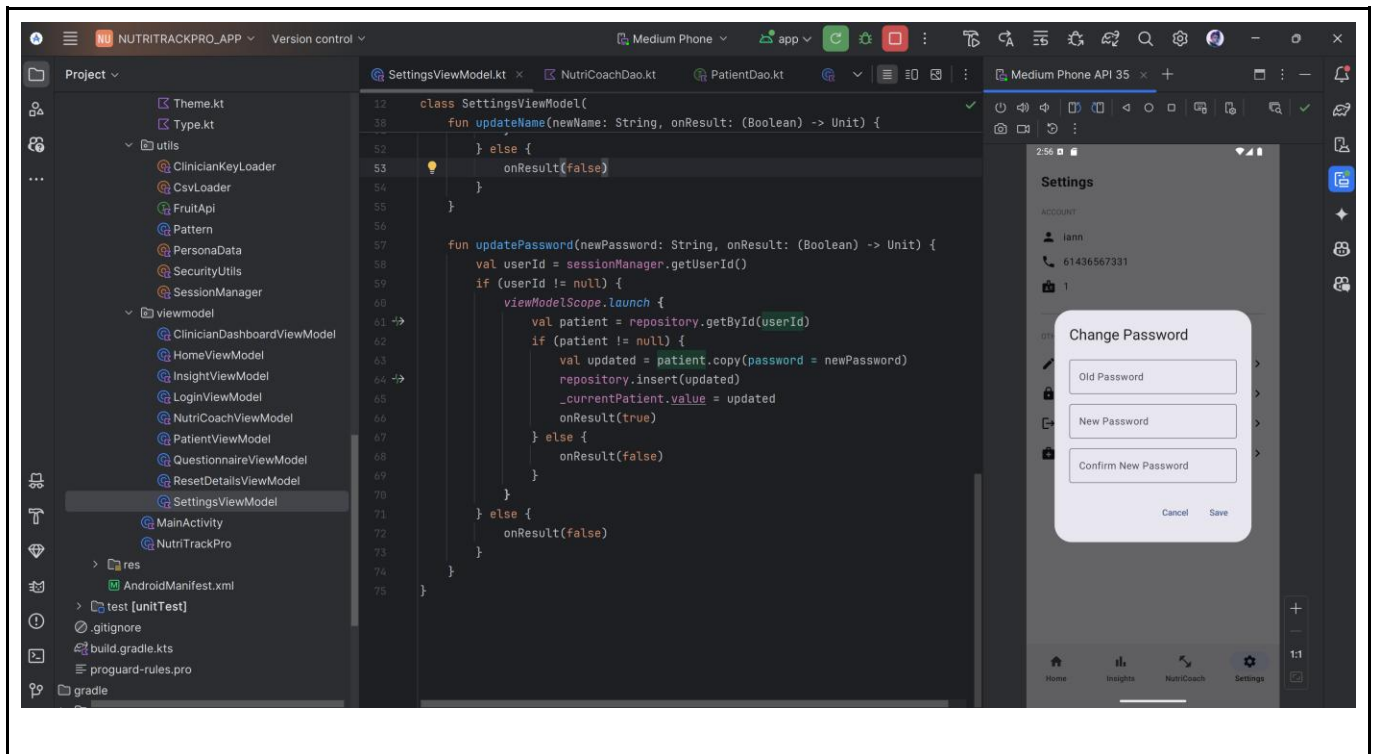
3. Feature Description (50-75 words) *(The role of the feature and how it works)*

This feature powers the Settings screen by showing the logged-in user's details and allowing them to update their name or password securely. It uses session data to load the correct user and reflect changes instantly. Giving users control to edit their own info without admin help, offering a smoother, self-service experience for account management.

3. Feature Screenshots (max 2)



Name: [IAN LEONG ZHENG YAN] **Student ID:** [34423680] **Lab:** [TUTORIAL 06]



1. Feature Name

Hashing User Password

2. Path to the Feature Class

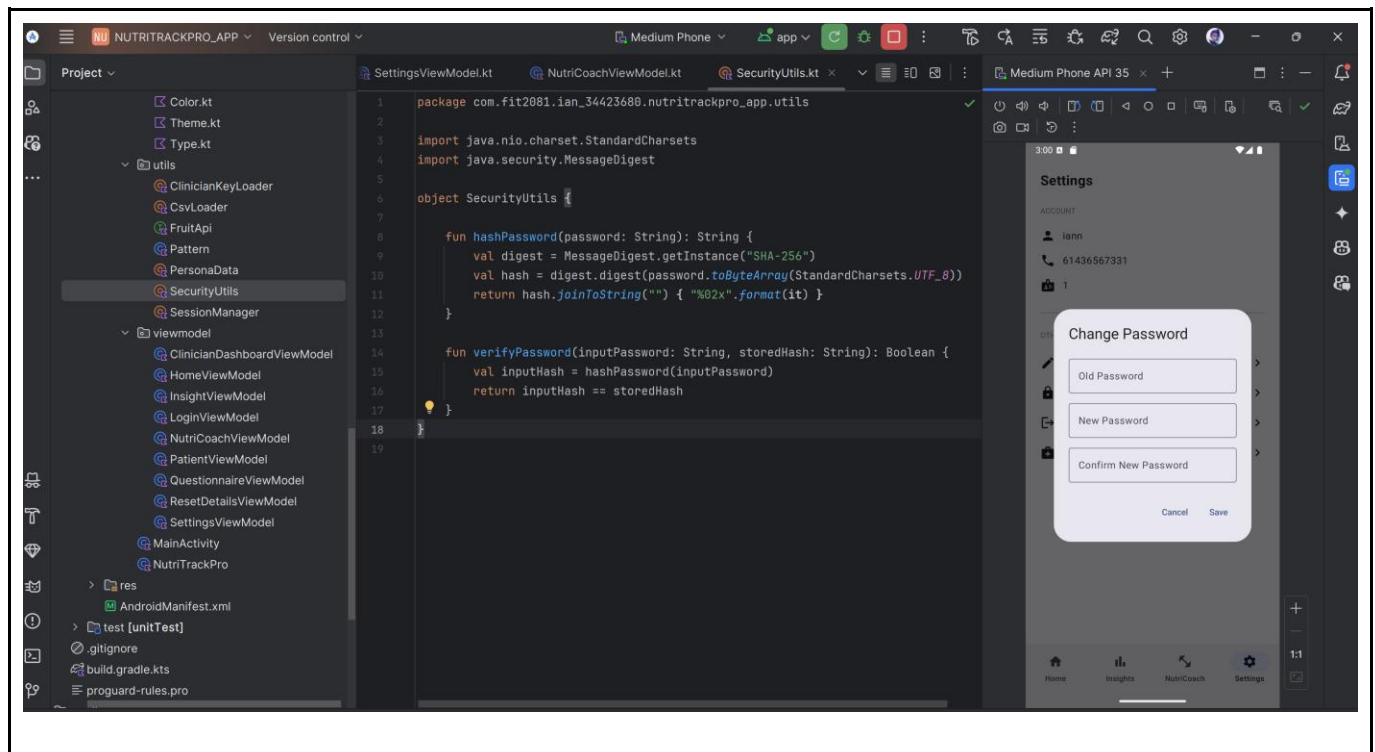
(e.g. `com/fit2081/a3_app/features/PatientMonitor.kt`)

```
import com.fit2081.ian_34423680.nutritrackpro_app.utils.SecurityUtils
com.fit2081.ian_34423680.nutritrackpro_app.repository.ClinicianKeyRepository.kt
com.fit2081.ian_34423680.nutritrackpro_app.repository.PatientRepository.kt
com.fit2081.ian_34423680.nutritrackpro_app.screen.SettingsScreen.kt
com.fit2081.ian_34423680.nutritrackpro_app.viewmodel.PatientViewModel.kt
```

3. Feature Description (50-75 words) *(The role of the feature and how it works)*

This feature securely handles password hashing and verification across the app. Instead of storing plain-text passwords, it uses SHA-256 to hash them, which protects user data even if the database is exposed. It also checks passwords by comparing hashes, not raw input. Implementing a proper security practice, making the app safer and more trustworthy as shown in the database.

3. Feature Screenshots (max 2)



Name: [IAN LEONG ZHENG YAN] Student ID: [34423680] Lab: [TUTORIAL 06]

The screenshot displays the Android Studio IDE interface for a project named 'NUTRITRACKPRO_APP'. The main editor shows the `SecurityUtils.kt` file with the following Kotlin code:

```
1 package com.fit2081.ian_34423680.nutritrackpro_app.utils
2
3 import java.nio.charset.StandardCharsets
4 import java.security.MessageDigest
5
6 object SecurityUtils {
7
8     fun hashPassword(password: String): String {
9         val digest = MessageDigest.getInstance("SHA-256")
10        val hash = digest.digest(password.toByteArray(StandardCharsets.UTF_8))
11        return hash.joinToString("") { "%02x".format(it) }
12    }
13
14    fun verifyPassword(inputPassword: String, storedHash: String): Boolean {
15        val inputHash = hashPassword(inputPassword)
16        return inputHash == storedHash
17    }
18 }
```

On the right, a mobile app preview shows a 'Settings' screen with fields for 'Name' (IAN), 'Phone' (6143656733), and 'Email' (ian34423680@gmail.com). Below the preview, the 'App Inspection' section shows the database inspector for the 'Medium Phone' instance, displaying the 'patients' table with the following data:

us...	sex	phone...	name	passwo...	Vegeta...	FruitHE...	Grainsa...	Wholeg...	Meatan...	Dairyan...	WaterH...	Unsat...	Saturat...
1	1	Female	6143656733	ian	10febaf0107787f90b75d6ae6e951acd3ff4b1390048a01924b5aece74541912					0.0	0.0	0.0	5.0
2	2	Female	6143656733	ian	10febaf0107787f90b75d6ae6e951acd3ff4b1390048a01924b5aece74541912					0.0	0.0	2.5	2.5
3	4	Male	6143656733		0.5	0.0	1.67	0.0	0.0	5.0	0.0	2.5	5.0
4	5	Male	6143656733		3.0	0.0	2.5	0.0	8.0	4.0	0.0	5.0	0.0

1. Feature Name

AnimatedVisibility ft NutriCoachScreen.kt

2. Path to the Feature Class

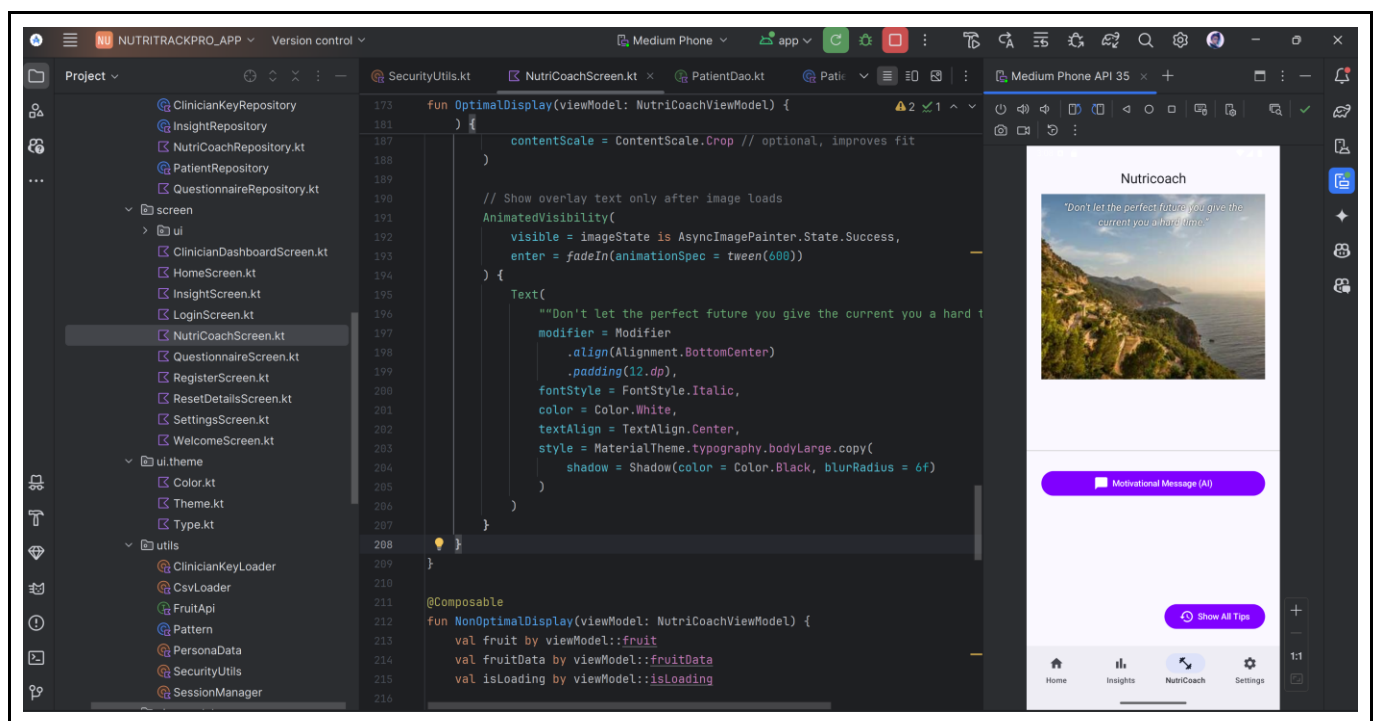
(e.g. `com/fit2081/a3_app/features/PatientMonitor.kt`)

`com.fit2081.ian_34423680.nutritrackpro_app.screen.NutriCoachScreen`

3. Feature Description (50-75 words) *(The role of the feature and how it works)*

This feature shows a motivational quote on top of a background image but only after the image has successfully loaded. The quote fades in smoothly for a nice visual effect. It's styled with italics and a subtle drop shadow to ensure it's readable against any background. This adds polish and emotional engagement, enhancing the user experience beyond basic UI.

3. Feature Screenshots (max 2)



1. Feature Name

Clinician Login with Hash Credentials and making it's own database

2. Path to the Feature Class

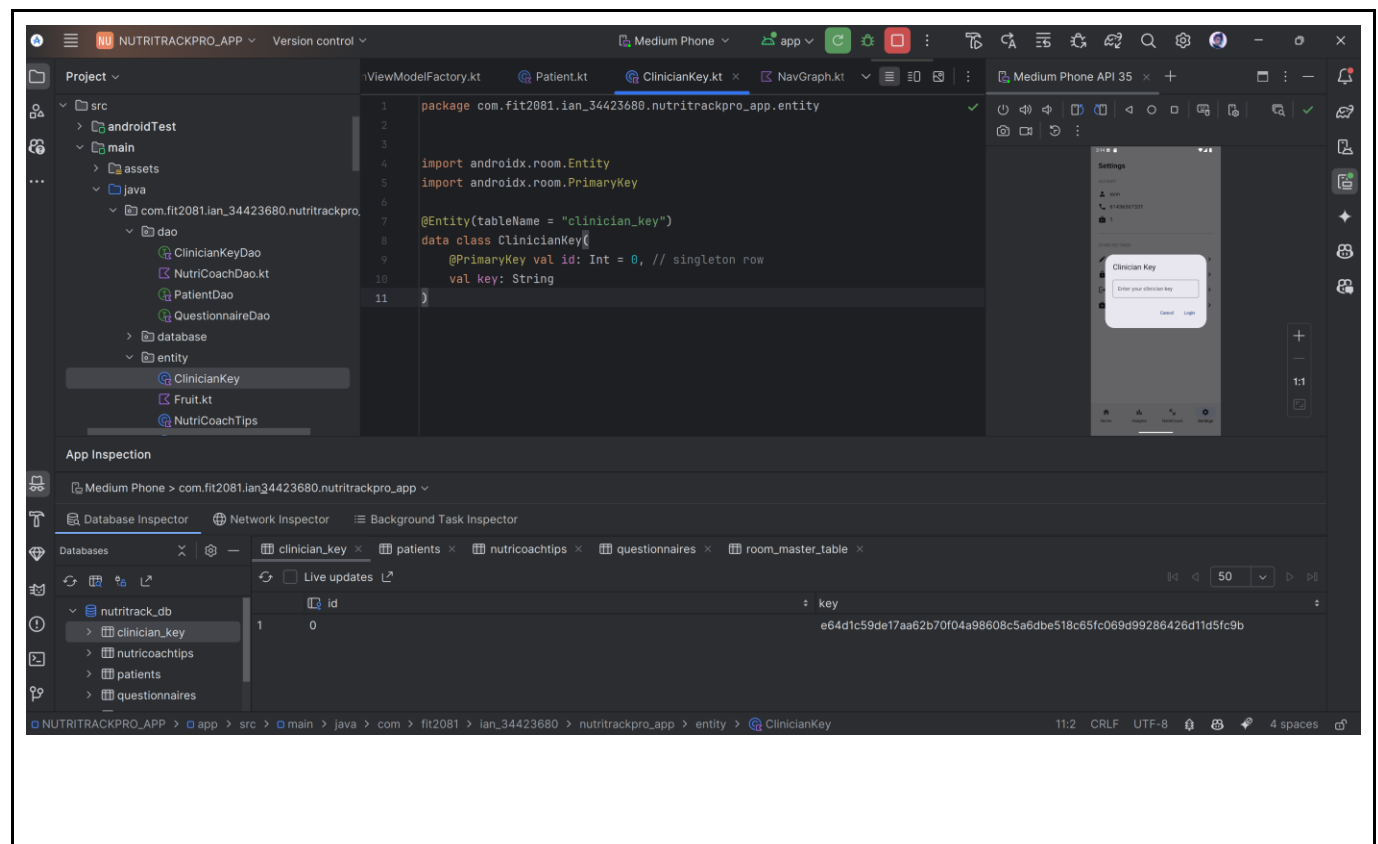
(e.g. `com/fit2081/a3_app/features/PatientMonitor.kt`)

`com.fit2081.ian_34423680.nutritrackpro_app.entity.ClinicianKey`

3. Feature Description (50-75 words) *(The role of the feature and how it works)*

This feature introduces a secure clinician login system using a dedicated database table and SHA-256 hashed passwords. It separates clinician access from patient data for better security and future scalability. By avoiding hardcoded keys and allowing dynamic clinician entries, it protects against credential leaks and prepares the app for future clinical users with unique logins to support real world use.

3. Feature Screenshots (max 2)



. Feature Name

Smart Login Redirection Based on Questionnaire Completion

2. Path to the Feature Class

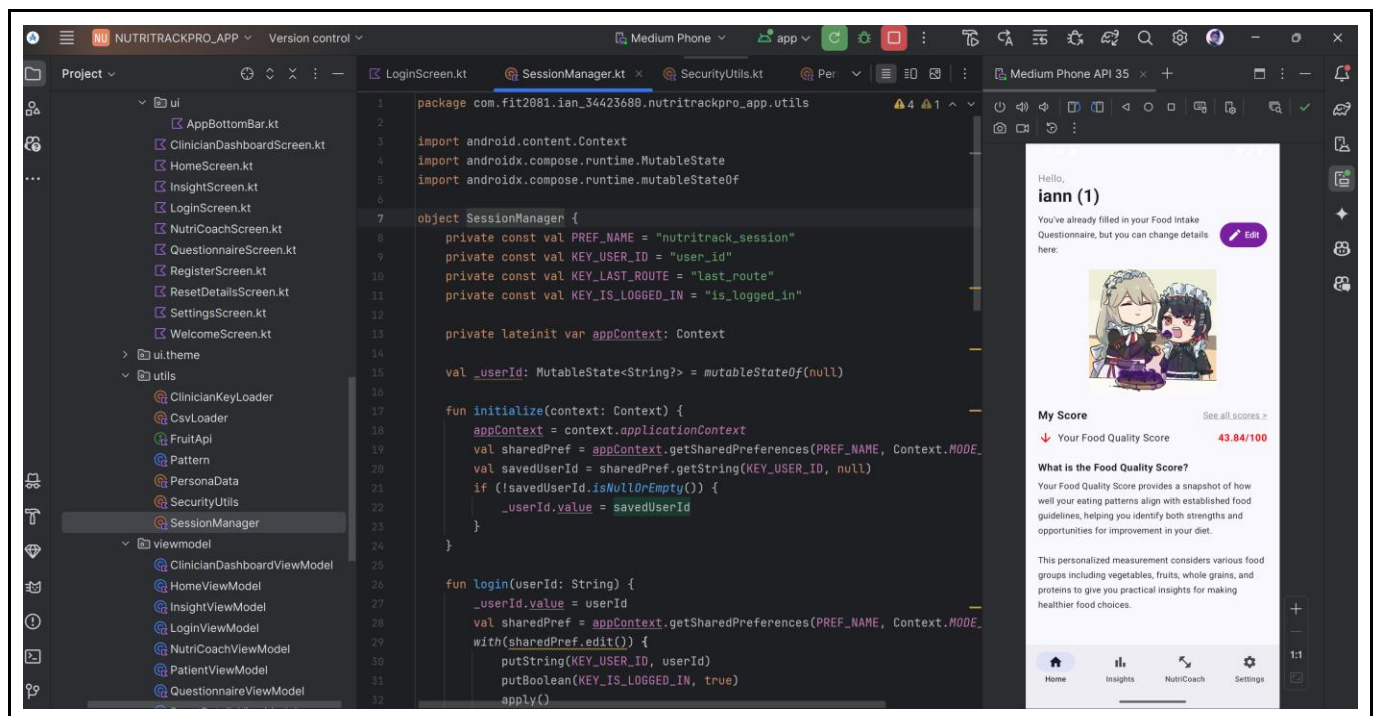
(e.g. `com.fit2081/a3_app/features/PatientMonitor.kt`)

`com.fit2081.ian_34423680.nutritrackpro_app.screen.WelcomeScreen.kt`
(also integrated in `LoginScreen.kt` and handled via `SessionManager.kt`)

3. Feature Description (50-75 words) *(The role of the feature and how it works)*

This feature uses `SharedPreferences` (via `SessionManager`) to persist user login state and questionnaire progress across sessions. When the app launches, the Welcome screen checks whether a user is logged in and if they've completed the questionnaire. It intelligently redirects them to either the Home or Questionnaire screen based on this state skipping unnecessary screens.

3. Feature Screenshots (max 2)



Name: [IAN LEONG ZHENG YAN] **Student ID:** [34423680] **Lab:** [TUTORIAL 06]

. Feature Name

Clinician Login & AI Insights Using JSON and SHA-256

2. Path to the Feature Class

(e.g. `com.fit2081/a3_app/features/PatientMonitor.kt`)

`com.fit2081.ian_34423680.nutritrackpro_app.viewmodel.ClinicianDashboardViewModel`

3. Feature Description (50-75 words) (The role of the feature and how it works)

This feature adds a secure and scalable Clinician Login system using a hashed passphrase and a separate Room database. Once authenticated, clinicians can access HEIFA averages and trigger a GenAI insight generator. The AI returns structured clinical patterns in **JSON format**, which the ViewModel parses and displays as motivational insights.

3. Feature Screenshots (max 2)

